

DER VANG

Research Scientist || Analytical Chemist || Spectroscopist || Data Scientist || Applied Microbiologist || U.S. Citizen

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EDUCATION

Ph.D.	2019-2024	University of Cincinnati , Cincinnati, OH Chemistry Dissertation: "Intrinsic surface-enhanced Raman scattering (SERS) as a tool for biological analysis" Advisor: Prof. Pietro Strobbia	GPA: 3.88/4
B.S.	2015-2019	Winona State University , Winona, MN ACS Chemistry Major & Business Administration Minor Study Abroad: Chung Ang University, Seoul, South Korea, June-July 2017	GPA: 3.68/4 (Cum Laude Honors)

RESEARCH EXPERIENCE

Research Assistant, University of Cincinnati, Dept. of Chemistry, Cincinnati, OH

Advisor: Prof. Pietro Strobbia (Dec 2021- Dec 2024)

SERS substrates for biological applications

- Investigated nanostars morphology and kinetics to enhance sensitivity for different laser emissions via TEM & UV-vis
- Established methods to fabricate substrates and characterization with reflectance approaches to ensure reproducibility
- Led to a first-author publication, one oral, and multiple poster presentations

SERS and machine learning for identifying and classifying exosomes

- Demonstrated an approach to analyze cancerous and healthy exosomes with SERS and machine learning for non-invasive cancer screening and improving human comprehension of machine learning models
- Successfully utilized machine learning to identify trends in SERS spectra of commercial and clinical exosomes (BSL-1,2)
- Resulted in multiple poster presentations and a poster award at SciX 2023

SERS and machine learning for monitoring bacteria, a collaboration with Tom Cambren at Procter & Gamble Co.

- Influenced the approach to monitor bacteria (BSL-1) samples with SERS and machine learning to uncover the mechanism of action of different antibacterial agents
- Produced highly informative SERS spectra that multivariate analysis was able to uncover differences among the treatments
- Established a cost-effective and faster approach, reducing the dependency on mass spectrometry and genome sequencing
- Resulted in multiple proceedings and an oral presentation at SPIE 2024

Research Assistant, University of Cincinnati, Dept. of Chemistry, Cincinnati, OH

Advisor: Prof. Peng Zhang (Jan 2020 – Dec 2021)

Nanoparticles and formulations to treat topical infectious diseases via photodynamic therapy

- Influenced the synthesis of hybrid nanoparticle photosensitizers for treating topical diseases
- Demonstrated the result of dye loading efficiency into nanoparticles and efficacy in treating *T. rubrum* (BSL-2)

Research Student, Winona State University, Dept. of Chemistry, Winona, MN

Advisor: Prof. Joseph K. West (Aug 2016-May 2019)

Developed a greener synthetic method for phosphine sulfides and selenides.

- Established a greener and faster protocol to oxidize phosphines, avoiding traditional reflux and hazardous solvents
- Leveraged a microwave reactor to decrease reaction time, and improved the purity and yield of 17 different targets
- Led to multiple poster presentations and one oral presentation.

Synthesis and application of an Air-stable Trifluoromethylating Agent.

- Created a robust protocol to synthesize a trifluoromethylation agent in less than 3 h compared to 2.5 days
- Demonstrated how an NMR reaction scale can be used to test the agent's efficacy before scaling up the synthesis
- Resulted in a coauthored publication

PROFESSIONAL EXPERIENCE

Apr 2026 – current	Associate Data Manager, Medpace , Cincinnati, OH Coordinate project timeline for Data Management activities. Validate and review documents of database set-up and data transfer. Oversight study activities and mentor Data Coordinators.
Jan 2025 – Mar 2026	Clinical Data Coordinator III, Medpace , Cincinnati, OH Validate database designs, reconcile data from multiple sources, and resolve data conflicts in collaboration with global research teams. Analyze data trends, and identify issues.
Aug 2024 – Dec 2024 Aug 2019 – Aug 2022	Teaching Assistant, University of Cincinnati, Dept. of Chemistry , Cincinnati, OH General Chemistry I Lab (CHEM 1040L); Organic Chemistry I & II Lab (CHEM 2040L & 2041L); Instrumental Analysis Lab (CHEM 3030L); Introduction to Biochemistry (CHEM 3040)
Oct 2018 – Apr 2019	Associate Quality Assurance Lab Technician, The Watkins Co. , Winona, MN Conduct analytical (e.g., FT-IR, pH, melting temperature), sensory (e.g., sight, smell, taste) , and biological analyses on raw materials and manufactured goods. Maintained a safe and clean laboratory environment.

Aug 2016 – Apr 2019	Lab/ Teaching Assistant, Winona State University, Department of Chemistry , Winona, MN General, Organic & Biochemistry Lab (CHEM 210), Principles of Chemistry I & II Lab (CHEM 212 & 213); Principles of Organic Chemistry II Lab (CHEM 351)
Mar 2013 – Oct 2018	Supervisor, Dairy Queen , Winona, MN Provided leadership, mentorship, and training for new employees. Enhanced operations and productivity. Managed daily cash intake and tabulating profits. Served over 50 customers per shift.
Mar 2014 – Oct 2014	Receptionist, Fantastic Sams , Winona, MN Check-in customers. Scheduled appointments. Cleaned up stations.

SKILLS

Material Characterization Techniques – NMR, FTIR, Rama, SERS, Atomic Absorption, TEM, SEM, UV-vis, Fluorescence, DLS, NTA, SEC, HPLC, GC-MS, ICP-MS

Special Lab Skills – Titration, Dilution, Schlenk line, Bacteria culture, Recrystallization

Nanomaterial Synthesis – Au and Ag nanospheres, (silver-coated) gold nanostar, silica nanoparticles

Computer Skills – Matlab (Intermediate), IgorPro (Intermediate), Microsoft Office (e.g., Excel) (Pro)

PUBLICATIONS (8 scholarly, 3 proceedings, 1 perspective, & 1 bio-archive)

- Giordano, A.; Franqui-Rios, S.; Quarin, S.; **Yang, D.**; Austin, D.; Doyle, A.; Baldwin, L.; Strobbia, P.; Rao, R. "Fabrication of SERS Substrates Using Silver-Coated Gold Nanostars for Chemical Sensing: A Multiobjective Bayesian Optimization Approach". *ACS Applied Nano Materials* **2025**, 8 (23), 11930–11939.
- Yang, D.**; Pahren, J.; Duderstadt, E.; Alvarez, F. J.; Caserta, J.; Cambron, T.; Strobbia, P. "Surface-enhanced Raman Spectroscopy (SERS) and Multivariate Analysis for Elucidating Mechanisms of Action in Antibacterial Agents". *ACS Sensors* **2025**, 10 (4), 2689–2698.
- Quarin, S. M.; **Yang, D.**; Dima, R. I.; Stan, G.; Strobbia, P. "AI in SERS Biosensing: from discriminative to generative". *NPJ Biosensing* **2025**, 2 (1), 9.
- Yang, D.**; Kelly, M. S.; Sharma, M.; Sheokand, M.; Esfandiari, L.; Dima, R. I.; Strobbia, P. "Machine Learning Approaches in Label-Free Small Extracellular Vesicles Analysis with Surface-Enhanced Raman Scattering (SERS) for Cancer Diagnostics". *bioRxiv* **2024**.02.19.581099.
- Naher, L.; Quarin, S.; **Yang, D.**; Strobbia, P. "Lyophilizing SERS Biosensors to Enable Use in Point-of-Care Diagnostics". *Analytical Methods* **2024**, 6 (44), 7613–7623.
- Han, D.; **Yang, D.**; Adehinmoye, A.; Kissell, L.; Strobbia, P.; Steckl, A. J. "RAPID POINT-OF-CARE LFA DIAGNOSTICS FOR ORAL HEALTH USING OPTICAL EXCITATION/DETECTION METHODS". *Hilton Head* **2024**.
- Yang, D.**; Pahren, J.; Cambron, T.; Strobbia, P. "Label-free surface-enhanced Raman scattering (SERS) and machine learning for biological analysis". *Proc. of SPIE* **2024**, 13059, 130590C.
- Kissell, L. N.; Han, D.; **Yang, D.**; Cikanek, A. W. R.; Steckl, A. J.; Strobbia, P. "Improved Point-of-Care Detection of *P. gingivalis* Using Optimized Surface-Enhanced Raman Scattering in Lateral Flow Assays". *Sens. Diagn.* **2024**, 3, 839–849.
- Kissell, L. N.; Liu, H.; Sheokand, M.; **Yang, D.**; Kachroo, P.; Strobbia, P. "Direct Detection of Tobacco Mosaic Virus in Infected Plants with SERS-Sensing Hydrogels". *ACS Sensors* **2023**, 9 (1), 514–523.
- Confederat, S.; Lee, S.; **Yang, D.**; Soulias, D.; Marcuccio, F.; Peace, T. I.; Edwards, M. A.; Strobbia, P.; Samanta, D.; Wält C.; Actis, P. "Nanopore Sensors for Enhanced Detection of Nanoparticles". *Small* **2023**, 2305186.
- Pahren, J.; **Yang, D.**; Caserta, J.; Strobbia, P.; Cambron, T. "Online characterization of bacteria culture using Raman and SERS". *Proc. of SPIE* **2023**, 12548, 125480E.
- Yang, D.**; Strobbia, P. "Analysis of Nanostar Reshaping Kinetics for Optimal Substrate Fabrication". *Applied Spectroscopy* **2022**, 77 (3), 270–280.
- West, J. K.; **Yang, D.** "(Ph₃P)₃CuCF₃: Preparation and Application of an Air-stable Trifluoromethylating Agent". *J. Chem. Educ.* **2022**, 99 (10), 3573–3577.

ORAL PRESENTATIONS

- Der Yang**, Maria S. Kelly, Manisha Sheokand, Manju Sharma, Leyla Esfandiari, Ruxandra I. Dima, Pietro Strobbia; "Machine Learning Methods in Exosomes Analysis with Surface-Enhanced Raman Spectroscopy (SERS) for Label-Free Cancer Detection" **SciX 2024 Conference**, Rayleigh Convention Center, Oct 24, 2024.
- Der Yang**, Maria S. Kelly, Ruxandra I. Dima, Jonathan Pahren, Tom Cambron, Pietro Strobbia; "Label-free surface-enhanced Raman scattering (SERS) and machine learning for biological analysis" Smart Biomedical and Physiological Sensor Technology XXI, part of **SPIE Defense + Commercial Sensing (DCS) 2024**, Gaylord National Resort & Convention Center, Apr 22, 2024.
- Der Yang** and Pietro Strobbia; "Analysis of Nanostar Reshaping Kinetics for Optimal Substrate Fabrication" **SciX 2022 Conference**, Northern Kentucky Convention Center, Oct 3, 2022.
- Der Yang** and Joseph Kent West; "“GREENER” METHOD FOR THE SYNTHESIS OF PHOSPHINE SELENIDES" **Winchell Undergraduate Research Symposium**, University of St. Thomas, April 27, 2019.

POSTER PRESENTATIONS

- **Der Vang**, Maria S. Kelly, Manisha Sheokand, Manju Sharma, Leyla Esfandiari, Ruxandra I. Dima, Pietro Strobbia; "Optimizing Label-Free Exosome Analysis with Surface-Enhanced Raman Scattering (SERS) and Machine Learning for Non-Invasive Cancer Diagnostics" **SciX 2023 Conference**, Nugget Casino Resort, Oct 8 & 10, 2023.
- **Der Vang** & Pietro Strobbia; "Analysis of Nanostar Reshaping Kinetics for Optimal Substrate Fabrication" **Midwest Universities Analytical Chemistry Conference (MUACC)**, University of Cincinnati, Oct 28, 2022.
- **Der Vang** & Joseph K. West, Ph.D.; "Synthesis and Application of an Air-stable Trifluoromethylating Agent as an Advanced Inorganic Chemistry Laboratory Experiment" **Ramaley Research Celebration**, Winona State University, April 17, 2019.
- **Der Vang**, Ashley B. Olichwier, & Joseph K. West; "Green, facile method for the synthesis of phosphine sulfides and selenides" **ACS National Meeting & Exposition**, Boston Convention & Exhibition Center. August 19, 2018.
- **Der Vang** & Joseph K. West, Ph.D.; "'Greener' method for the synthesis of phosphine selenides" **Ramaley Research Celebration**, Winona State University, April 11, 2018.
- **Der Vang** & Joseph K. West, Ph.D.; "'Green' protocol for the oxidation of triarylphosphines by sulfur and selenium" **Ramaley Research Celebration**, Winona State University, April 2017.

FELLOWSHIPS, AWARDS, AND HONORS

2024	Dr. William R. Heineman Fellowship, University of Cincinnati, Department of Chemistry, for one graduate student per year pursuing research in analytical chemistry, with a preference given to interdisciplinary work, the 2 nd awardee
2023	Society of Applied Spectroscopy (SAS) Student Poster Award, 1 of 4 awardees among 46 poster presentations, SciX 2023 Conference, for Outstanding Spectroscopic Research By A Student
2023	URC Graduate Student Stipend and Research Cost Program for Faculty-Student Collaboration Award, University of Cincinnati, College of Research, competition among students from physical science & engineering colleges
2022-2023	Institute of Research in Sensing (IRiS) Graduate Research Fellowship Award, University of Cincinnati
2019-2021	Orchin Research Associate Award, University of Cincinnati, Department of Chemistry, for an incoming graduate student with outstanding accomplishment and future research potential
2019	Cum Laude Honors, Winona State University
2016-2018	Academic & Research Awards at Winona State University, Department of Chemistry (Marilyn & Frederick Foss Scholarship 2018), (Winona State Student Research Grant 2017), (C.B. William Ng Chemistry Scholarship 2017, 2016), (Cecil W. Gronvall Memorial Scholarship 2016)
2014	Harvard Book Award, Winona Senior High School, Winona, MN, to one outstanding high school student

PROFESSIONAL AFFILIATIONS

Mar 2024 – Mar 2025	Society of Photo-Optical Instrumentation Engineers (SPIE)
Oct 2023 – Oct 2025	Coblentz Society
Nov 2022 – current	Society for Applied Spectroscopy (SAS)
Jun 2022 – current	American Chemical Society (ACS) Division of Analytical Chemistry
Jun 2022 – Jun 2025	American Chemical Society (ACS) Division of Chemical Information, Division of Cellulose & Renewable Materials, and Division of Catalysis Science & Tech
Jun 2018 – current	American Chemical Society (ACS)

MENTORSHIP, VOLUNTEERISM, OUTREACH, & SERVICE

Undergraduate Mentorship – 3 University of Cincinnati Students (Spring 2023 & 2024), 1 Virginia State University (Summer 2021)

Volunteerism at Univ. of Cincinnati – Showcase Reviewer, Undergraduate Scholarly Showcase (2023, 2024), Bearcat Pantry Volunteer Ambassador, helping with food insecurities (2023-2024), Recruiter for Dept. of Chem., Fall 2023 ACS Conference, Graduate Recruitment Fair

Outreach & Service – International Day of Light (2024, 2025, 2026) at Cincinnati Museum Center, OH, National Chemistry Week (2023) at Cincinnati Museum Center, OH

REFERENCES

Prof. Pietro Strobbia Ph.D. Dissertation Advisor Department of Chemistry University of Cincinnati, OH strobbpo@ucmail.uc.edu	Prof. Ryan White Dissertation Committee Member Department of Chemistry University of Cincinnati, OH ryan.white@uc.edu	Tom Cambron Collaborator Procter & Gamble Co Mason, OH cambron.rt@pg.com
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HOBBIES

Traveling, gardening, sewing, DIY crafts, baking, and spending time with loved ones.